

WEBSITE TRAFFIC ANALYSIS

Phase 3: Development Part 1 - Loading and Preprocessing the Dataset

Introduction

In this phase of the project, we embark on building a Website traffic analysis utilizing IBM Cognos for visualization. The primary focus here is on loading and preprocessing the website traffic dataset to ensure it is ready for analysis.

Project Objectives

The aim of this project is to enable how to utilize our Data Visualization and Data Analytics skills.

1. Understanding daily website visitor trends.
2. Identifying peak traffic periods.
3. Analyzing the impact of marketing campaigns or other external events on website traffic.
4. Monitoring user engagement and interaction with the website.
5. Assessing the effectiveness of landing pages and content.

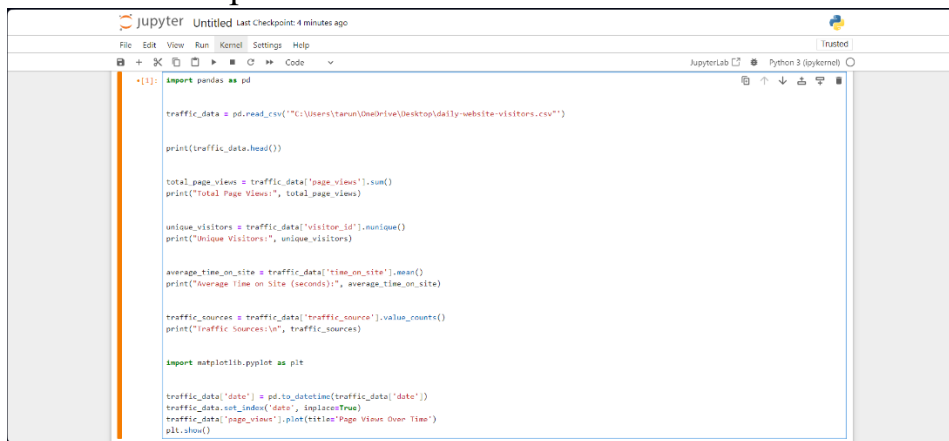
Analysis Objectives

The objectives of website traffic analysis can vary based on the specific goals and needs of a business or organization.

Data Source: We obtained our data from Kaggle, a reputable platform for datasets. The dataset link is provided here: [Daily Website Visitors Dataset](#).

Data Loading: To begin the analysis, we loaded the website traffic dataset into our IBM Cognos environment. This step involved using the necessary functions

and tools to import the data.

A screenshot of a Jupyter Notebook interface. The top bar shows 'jupyter' and 'Untitled' with a timestamp 'Last Checkpoint: 4 minutes ago'. Below the menu bar (File, Edit, View, Run, Kernel, Settings, Help), there's a toolbar with icons for file operations and a status bar indicating 'JupyterLab', 'Python 3 (ipykernel)', and 'Trusted'. The main code area contains the following Python code:

```
import pandas as pd

traffic_data = pd.read_csv("C:\\Users\\tarun\\OneDrive\\Desktop\\daily-website-visitors.csv")

print(traffic_data.head())

total_page_views = traffic_data['page_views'].sum()
print("Total Page Views:", total_page_views)

unique_visitors = traffic_data['visitor_id'].nunique()
print("Unique Visitors:", unique_visitors)

average_time_on_site = traffic_data['time_on_site'].mean()
print("Average Time on Site (seconds):", average_time_on_site)

traffic_source = traffic_data['traffic_source'].value_counts()
print("Traffic Sources:\n", traffic_source)

import matplotlib.pyplot as plt

traffic_data['date'] = pd.to_datetime(traffic_data['date'])
traffic_data.set_index('date', inplace=True)
traffic_data['page_views'].plot(title='Page Views Over Time')
plt.show()
```

Data Preprocessing:

Data preprocessing is a critical step to ensure the data's accuracy and reliability. It involves several key aspects:

- ❖ **Handling Missing Values:** We meticulously dealt with missing data points. This might include imputation, removal of incomplete records, or other strategies based on the nature of the data.

- ❖ **Data Transformation:** Depending on the analysis objectives, we performed data transformations such as normalizing values, aggregating data over specific time periods, or converting data types to match the analytical requirements.

- ❖ **Data Cleaning:** Data cleaning involved removing outliers, duplicate entries, or any irrelevant information that could skew the analysis.

- ❖ **Data Verification:** To ensure the dataset's accuracy and reliability, we conducted verification steps, including cross-checking data with authoritative sources, validating data against known statistics, and conducting internal

consistency checks.

IBM Cognos Analytics

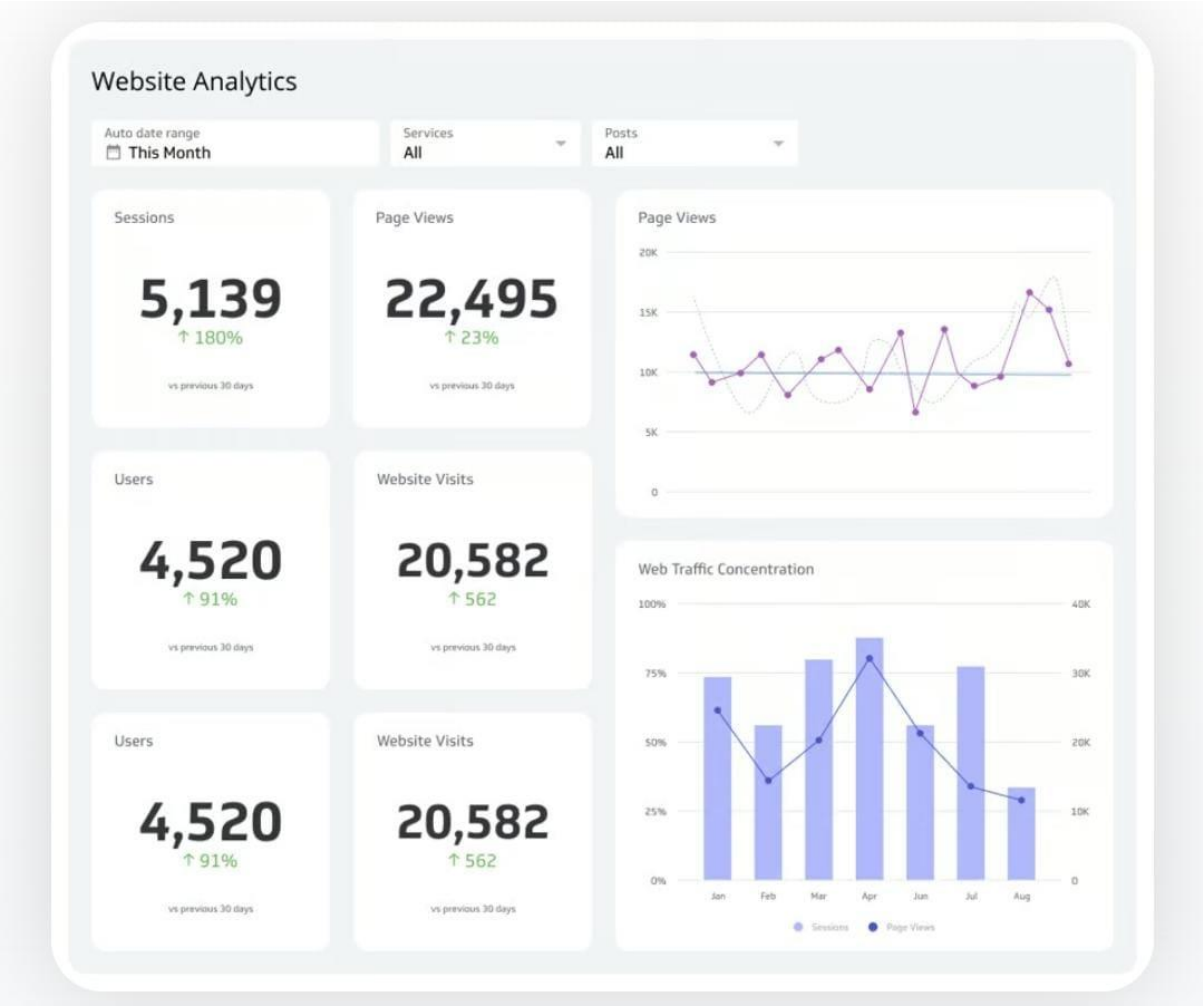
New data module

Grid

Relationships

Custom tables

Row Id	Row	Day	Day.Of.Week	Date	Page.Loads	Unique.Visits
1	1	Sunday	1	9/14/14	2,146	1,582
2	2	Monday	2	9/15/14	3,621	2,528
3	3	Tuesday	3	9/16/14	3,698	2,630
4	4	Wednesday	4	9/17/14	3,667	2,614
5	5	Thursday	5	9/18/14	3,316	2,366
6	6	Friday	6	9/19/14	2,815	1,863
7	7	Saturday	7	9/20/14	1,658	1,118
8	8	Sunday	1	9/21/14	2,288	1,656
9	9	Monday	2	9/22/14	3,638	2,586
10	10	Tuesday	3	9/23/14	4,462	3,257
11	11	Wednesday	4	9/24/14	4,414	3,175
12	12	Thursday	5	9/25/14	4,315	3,029
13	13	Friday	6	9/26/14	3,323	2,249



CONCULSION:

This part of the development phase has greatly focused on laying the foundations for the website traffic analysis. The dataset has been successfully loaded into the IBM COGNOS, preprocessed , and verified for accuracy.